



SEMESTER 2 EXAMINATIONS 2024/2025

MODULE: CSC1021 - Operating Systems

PROGRAMME(S):

CASE	BSc Computer Apps (Sft.Eng.)
COMSCI	BSc Computer Science
ECSA	Study Abroad Program EC
ECSAO	Study Abroad Program EC

STAGE: 1,2

EXAMINER(S):

Graham Healy	(Internal)	(Ext: 7364)
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TIME ALLOWED: 2 Hours

INSTRUCTIONS: Answer all questions.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.

There are no additional requirements for this paper.

QUESTION 1**[TOTAL MARKS: 25]****Q 1(a)****[8 Marks]**

Describe the Linux process life-cycle, indicating the state transitions that can occur for a process e.g. *new* state to a *ready* state. In your answer, explain the role that the operating system scheduler has in these state transitions.

Q 1(b)**[4 Marks]**

Briefly explain what the C function *exec/p()* does on Linux.

Q 1(c)**[8 Marks]**

Explain the memory layout of the C program below. In your answer, state what will be stored in each of the following memory sections when the program shown below is executed: heap, text, initialised data, uninitialised data and stack.

C program for Q1(c)

```
00: #include <stdio.h>
01: #include <stdlib.h>
02:
03: int a=2;
04: int b;
05:
06: int main(int argc, char **argv)
07: {
08:     int *vals;
09:     int i;
10:
11:     vals = (int *)malloc(sizeof(int)*10);
12:
13:     for (i = 0; i < 10; i++)
14:         vals[i] = i;
15:
16:     return 0;
17: }
```

Q 1(d)**[5 Marks]**

Explain why an executable program compiled on one system may not be executable on another system.

[End of Question 1]

QUESTION 2

[TOTAL MARKS: 25]

Q 2(a)

[7 Marks]

Using Amdahl's law, calculate the speedup of a program with a 25% serial proportion for 5 processors. Assuming there is an unlimited number of processors available, what's the maximum speedup that could be achieved?

Q 2(b)

[8 Marks]

The following C code demonstrates the operation of the compare-and-swap instruction that executes atomically. Provide a code example that shows how the compare-and-swap instruction could be used to implement an atomic integer variable i.e. show you would implement **`void increment(atomic_int *v)`**

```
int compare_and_swap( int *value, int expected, int new_value )
{
    int t = *value;

    if (*value == expected)
        *value = new_value;

    return t;
}
```

Q 2(c)

[10 Marks]

In the table below, the shortest-remaining-time-first scheduling is used to schedule processes for execution on a single-core CPU. Show the execution order of the processes (e.g. using a Gantt Chart) and calculate the average waiting time of the processes.

Process	Arrival Time	Burst Time
P1	0	8
P2	1	4
P3	2	9
P4	3	5

[End of Question 2]

QUESTION 3

[TOTAL MARKS: 25]

Q 3(a)

[7 Marks]

Using an example of an integer variable *counter*, where 2 threads execute *counter++* concurrently, explain what a *race condition*. In your answer, describe how a programmer would prevent a *race condition*.

Q 3(b)**[7 Marks]**

Explain the *wait()* and *signal()* operations of the semaphore synchronisation primitive. In your answer, explain how busy waiting can be avoided in the implementation of the *wait()* operation.

Q 3(c)**[11 Marks]**

The following program will end up in a deadlocked state. Explain why, by discussing each of 4 necessary conditions for deadlock, and explain how each applies to the program below with reference to specific line numbers.

Python code for Q3(a)	
00: import threading 01: 02: lock1 = threading.Lock() 03: lock2 = threading.Lock() 04: 05: def thr1(): 06: while True: 07: lock1.acquire() 08: lock2.acquire() 09: print("CS: Thr1") 10: lock1.release() 11: lock2.release()	12: def thr2(): 13: while True: 14: lock2.acquire() 15: lock1.acquire() 16: print("CS: Thread2") 17: lock2.release() 18: lock1.release() 19: 20: def main(): 21: threading.Thread(target=thr1).start() 22: threading.Thread(target=thr2).start() 23: print("running ...") 24: 25: main()

[End of Question 3]

QUESTION 4**[TOTAL MARKS: 25]****Q 4(a)****[4 Marks]**

The following sequence of commands are run on Linux:

```
echo "123" > myfile1.txt  
ln myfile1.txt myfile3.txt  
ln myfile1.txt myfile2.txt  
echo "XYZ" > myfile2.txt  
echo "ABC" > myfile3.txt
```

What (if anything) is contained in myfile1.txt?

Q 4(b)**[6 Marks]**

Explain the difference between hard and soft links in the context of a filesystem. In your answer, explain why hard links are not allowed for directories.

Q 4(c)**[12 Marks]**

Paging is a memory management scheme.

- i) Explain how a *page table* works by using an example of a 32-bit virtual address.
- ii) Describe the implications of using a small versus a large frame/page size.

Q 4(d)**[3 Marks]**

Explain what the following Linux command does:

`chmod 711 myfile`

[End of Question 4]

[END OF EXAM]